

CHEHALIS, WA, USA

WMO: 720254

Lat: 46.683N

Lon: 122.983W

Elev: 178

StdP: 14.60

Time Zone: -8.00 (NAP)

Period: 07-19

WBAN: 00119

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	22.8	26.8	16.3	12.6	26.6	21.5	16.2	28.7	19.0	44.7	17.0	45.6	1.2	210	0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	25.3	90.0	68.8	85.6	66.8	81.5	65.2	69.9	85.7	68.1	83.1	66.2	79.3	6.6	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.8	89.3	74.3	63.0	86.7	72.7	61.2	81.2	70.5	33.8	86.1	32.4	83.1	30.8	79.8	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.1	13.0	11.4	DB	16.3	96.1	4.7	4.5	13.0	99.3	10.3	102.0	7.7	104.5	4.3	107.8	(4)
(5)			WB	16.0	73.7	4.4	1.7	12.9	75.0	10.3	76.0	7.8	77.0	4.6	78.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.2	41.5	42.4	45.9	48.7	56.0	60.4	65.7	66.4	61.1	52.4	45.2	40.1		
	DBStd	10.64	6.01	5.86	4.98	4.69	5.69	5.54	4.68	4.70	4.95	4.83	6.38	6.57		
	HDD50	1222	267	217	145	78	10	0	0	0	1	29	166	309		
	HDD65	4858	729	631	591	488	283	160	45	40	136	391	593	771		
	CDD50	2036	4	5	20	39	196	313	488	508	333	103	23	4		
	CDD65	196	0	0	0	0	4	23	67	83	19	0	0	0		
	CDH74	3158	0	0	3	17	199	395	1073	1122	341	8	0	0		
	CDH80	1148	0	0	0	2	44	141	417	448	96	0	0	0		
(14) Wind	WSAvg	4.0	4.6	4.6	4.8	4.5	3.5	3.4	3.3	3.0	3.2	3.8	4.9	4.8		
(15) Precipitation	PrecAvg	45.9	6.9	5.0	4.8	3.5	2.4	1.8	0.7	1.3	1.9	4.1	7.0	7.2		
	PrecMax	64.6	14.6	12.6	8.9	7.7	5.1	4.5	2.4	4.1	6.7	9.3	12.8	14.3		
	PrecMin	31.9	0.4	0.2	0.3	0.9	0.4	0.2	0.0	0.0	0.0	0.4	1.3	2.1		
	PrecStd	7.6	3.3	2.5	1.9	1.5	1.1	0.9	0.6	1.1	1.4	2.2	3.1	3.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.1	60.6	71.6	76.6	84.4	90.5	95.2	94.6	88.1	73.4	63.5	56.7		
		MCWB	51.7	51.9	56.2	59.9	64.0	70.2	70.1	68.5	68.0	60.1	58.4	54.1		
	2%	DB	54.2	55.5	63.5	68.2	79.5	83.8	89.8	90.0	82.1	67.7	59.1	54.1		
		MCWB	51.2	49.9	52.7	54.5	63.1	67.4	68.0	69.1	64.7	58.4	55.5	52.4		
	5%	DB	52.1	54.2	58.6	63.1	74.6	77.4	84.2	84.4	77.2	63.8	56.7	50.3		
		MCWB	50.3	49.9	50.5	52.1	60.8	62.9	67.2	67.2	62.9	56.8	53.6	48.9		
	10%	DB	49.9	51.5	54.8	58.8	69.6	72.9	80.7	81.0	72.9	61.1	54.5	48.1		
		MCWB	48.5	49.1	49.4	50.4	58.4	60.8	65.2	65.6	61.5	56.2	52.1	47.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.3	54.5	57.1	60.5	66.9	71.4	72.5	71.9	69.5	64.0	59.5	55.2		
		MCDB	56.2	57.4	64.8	74.2	81.3	88.3	90.5	89.0	82.1	69.9	61.9	56.3		
	2%	WB	52.3	52.6	54.2	56.4	63.7	67.7	69.8	69.5	66.5	60.7	56.6	52.3		
		MCDB	53.2	54.9	60.0	65.2	77.9	82.7	85.7	85.7	77.8	64.7	59.0	53.4		
	5%	WB	50.1	50.4	52.3	53.6	61.2	64.2	67.5	67.9	64.5	58.8	54.3	49.1		
		MCDB	51.0	52.9	57.3	60.5	72.9	75.8	83.1	83.1	73.7	62.2	56.2	49.7		
	10%	WB	48.0	48.2	50.4	51.7	58.6	61.9	65.5	66.1	62.7	56.9	52.6	46.6		
		MCDB	49.1	50.4	54.6	56.8	67.9	72.1	78.6	79.1	70.2	60.5	54.2	47.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.9	13.4	16.8	18.4	21.8	21.8	25.4	25.3	22.4	17.0	12.7	10.2		
		MCDBR	13.6	18.2	23.7	27.6	32.6	31.2	35.1	34.6	31.4	22.2	14.4	10.8		
	5% WB	MCWBR	9.7	11.7	13.4	15.1	16.7	15.3	15.5	15.0	15.4	13.4	10.1	8.2		
		MCWBR	10.6	14.5	19.9	23.7	29.3	28.9	32.0	31.3	26.5	17.4	12.0	9.2		
(40) Clear-Sky Solar Irradiance	taub		0.326	0.315	0.332	0.358	0.378	0.365	0.350	0.354	0.340	0.333	0.326	0.328		
		taud	2.482	2.545	2.509	2.402	2.362	2.422	2.486	2.476	2.522	2.539	2.506	2.424		
	Ebn at Noon		243	275	285	285	282	285	288	283	278	264	241	225		
		Edh at Noon	21	24	29	35	37	36	33	32	28	24	21	20		
(44) All-Sky Solar Radiation	RadAvg		334	601	883	1295	1641	1799	2012	1724	1280	734	394	277		
	RadStd		54	84	113	122	143	180	152	128	117	86	51	41		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+18

Nomenclature: See separate page